Unit

42 The transition metals

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Coloured compounds of transition metals
- Variable oxidation states of transition metals
- Transition metals and their compounds as catalysts

42.1 Introduction to transition metals

Across the second and third periods of the periodic table, there is a gradual change in properties, from alkali metals to halogens. The fourth period begins in the same way, with an alkali metal (potassium) and an alkaline earth metal (calcium). The next ten elements do not show the gradual change in properties like previous periods: they are all metals and remarkably similar to one another in their properties. They are called the first series of **transition metals**. Periods 5, 6 and 7 also contain transition metals (Fig. 42.1).

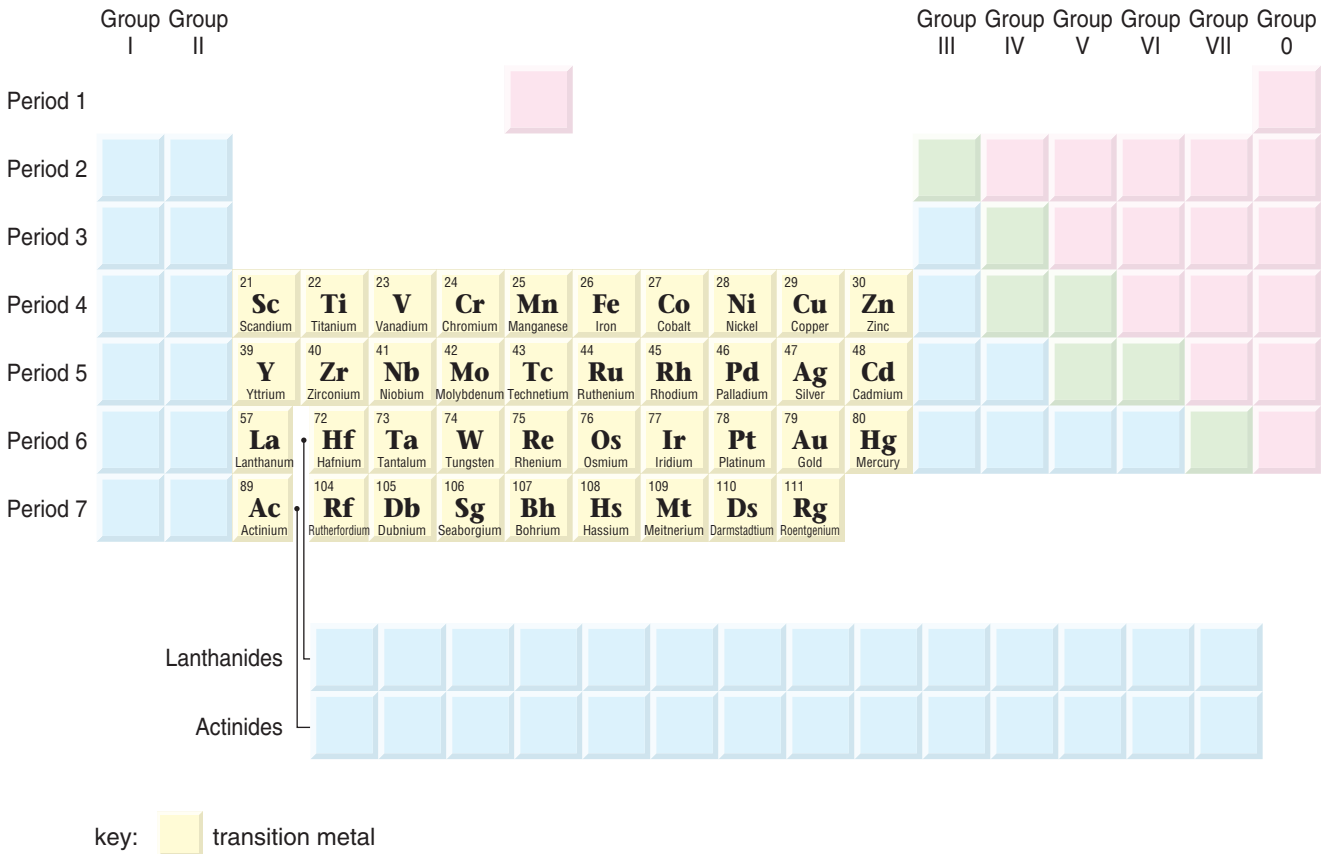


Fig. 42.1 A periodic table showing transition metals

transition metal 過渡金屬

Table 42.1 lists some physical properties of the first series of transition metals.

Table 42.1

Some physical properties of the first series of transition metals

Element	Symbol	Atomic number	Electronic arrangement of atom	Melting point (°C)	Boiling point (°C)	Density (g cm^{-3})	Atomic radius (pm)	Electrical conductivity
Scandium	Sc	21	2,8,9,2	1 541	2 836	3.0	162	3
Titanium	Ti	22	2,8,10,2	1 660	3 287	4.5	147	4
Vanadium	V	23	2,8,11,2	1 910	3 407	6.0	134	8
Chromium	Cr	24	2,8,13,1	1 907	2 671	7.2	128	11
Manganese	Mn	25	2,8,13,2	1 246	2 061	7.3	127	1
Iron	Fe	26	2,8,14,2	1 538	2 861	7.9	126	17
Cobalt	Co	27	2,8,15,2	1 495	2 927	8.9	125	24
Nickel	Ni	28	2,8,16,2	1 455	2 913	8.9	124	24
Copper	Cu	29	2,8,18,1	1 083	2 562	8.9	128	96
Zinc	Zn	30	2,8,18,2	420	907	7.1	134	27

Electrical conductivity is relative to an arbitrary value of 100 for silver which has the highest electrical conductivity of any element at room temperature.

The transition metals have certain common physical properties:

- they have high melting points;
- they have high densities;
- they are good conductors of electricity;
- they are hard and rigid, and so are useful as construction materials.

Do you know

Transition metals and plants

In the 1980s, scientists began to investigate the idea that plants might be used to extract certain metals. This work has resulted in phytomining (plant mining), where plants are used to extract transition metals from areas where these metals cannot be obtained economically by other methods.

In several states in the US, *Streptanthus polygaloides* are planted on nickel-rich soils. The plants take up nickel until it makes up as much as 1% of their dry mass. They are then harvested and burnt to ash, which is then smelted to produce the metal.

Fig. 42.2 shows uses of some transition metals.



Fig. 42.2 Uses of some transition metals

The physical properties of transition metals are noticeably different from those of the metals in the main groups. Some of the physical properties of chromium, iron and cobalt, together with those of potassium and calcium, are listed in table 42.2 to illustrate this point.

Table 42.2

Comparison of some physical properties between the transition metals and the main group metals

Physical properties	Transition metals			Metals in main groups	
	Cr	Fe	Co	K	Ca
Melting point (°C)	1 907	1 538	1 495	64	838
Boiling point (°C)	2 671	2 861	2 927	760	1 440
Density (g cm^{-3})	7.2	7.9	8.9	0.86	1.55
Atomic radius (pm)	128	126	125	227	197

N1 Refer to page T17.

Compared with metals in the main groups of the same period, the transition metals have higher melting points (Fig. 42.3), boiling points and density. Moreover, their atomic radii are smaller (Fig. 42.4). N1

AL 2009 Paper 1 Q9

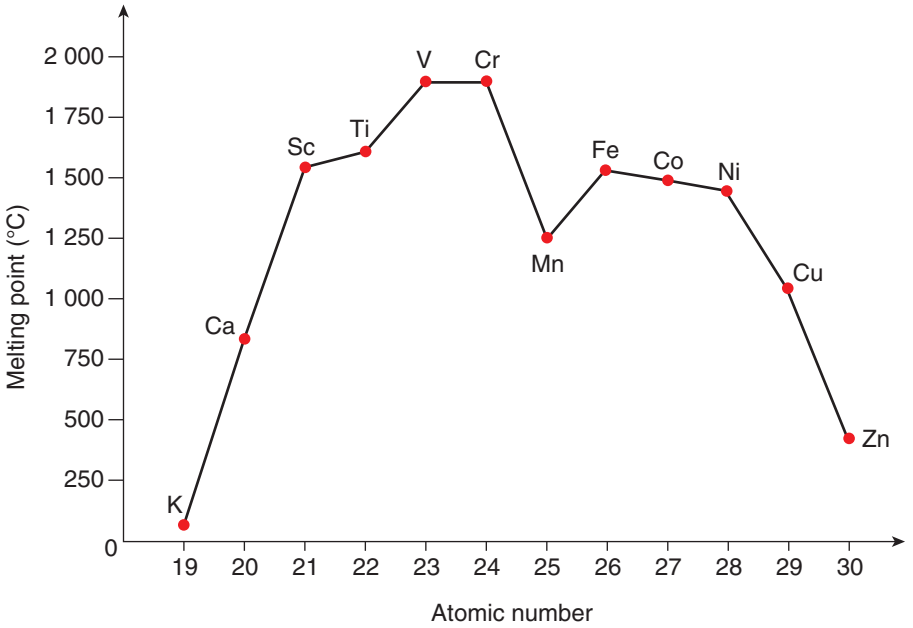


Fig. 42.3 The variation of melting points of the first 12 Period 4 elements

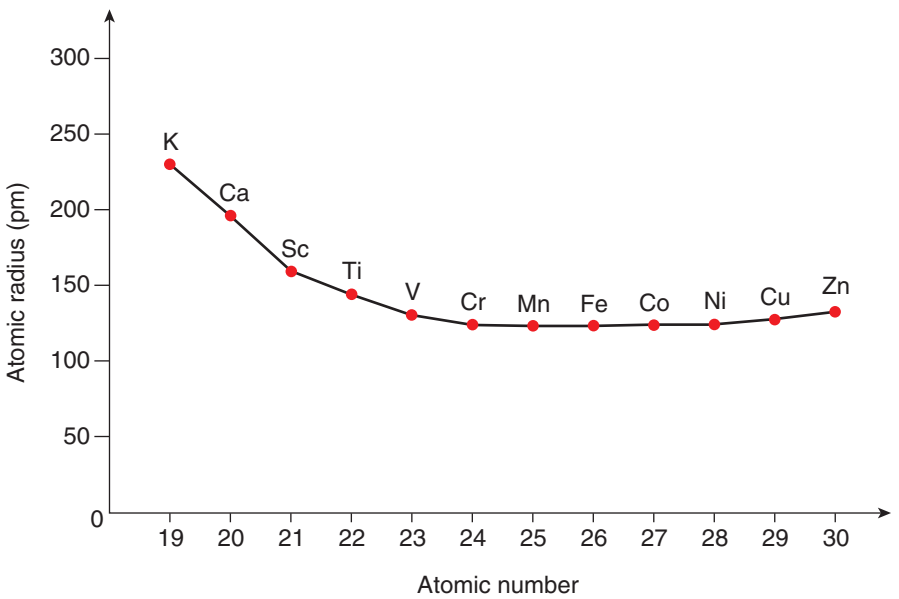


Fig. 42.4 The variation of atomic radii of the first 12 Period 4 elements

The chemical properties of transition metals are also very different from those of the metals in the main groups. We will discuss the following typical chemical properties of transition metals in this unit:

- formation of coloured compounds;
- existence in more than one **oxidation state** in their compounds; and
- catalytic properties of the elements and their compounds.

Elements on the edges of the block of transition metals, such as scandium and zinc, are not usually classified as transition metals. This will be discussed later in this unit.

DSE 2013 Paper 1A Q26
DSE 2014 Paper 1B Q11(a)
AL 2012 Paper 1 Q1(a)(iii)
AL 2009 Paper 1 Q9

42.2 Coloured compounds of transition metals

Visible light is a form of **electromagnetic radiation**. The visible light region consists of a spectrum with a range of wavelengths from approximately 400 nm to 700 nm (Fig. 42.5).

1 nm (nanometre) = 10^{-9} m

42.1
Inspecting compounds of transition metals.

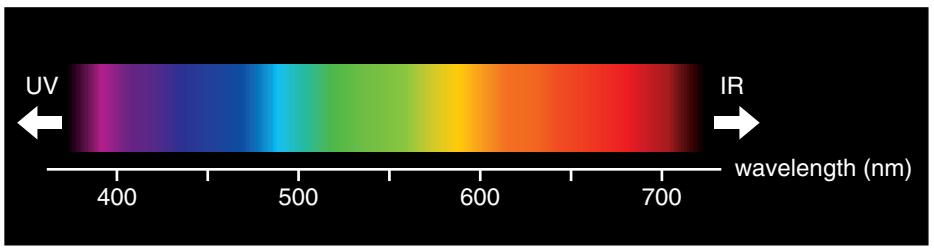


Fig. 42.5 The visible light region of the electromagnetic spectrum

When visible light hits a transparent solution, part of it may be absorbed and part of it may be transmitted. When sunlight passes through a copper(II) sulphate solution, copper(II) ions in the solution absorb light in the red, orange and yellow region of the spectrum. The remaining wavelengths in the light combine to give a blue colour as seen by our eyes.

oxidation state 氧化態　　electromagnetic radiation 電磁輻射

Fig. 42.6 gives an impression of what happens if sunlight passes through a copper(II) sulphate solution.

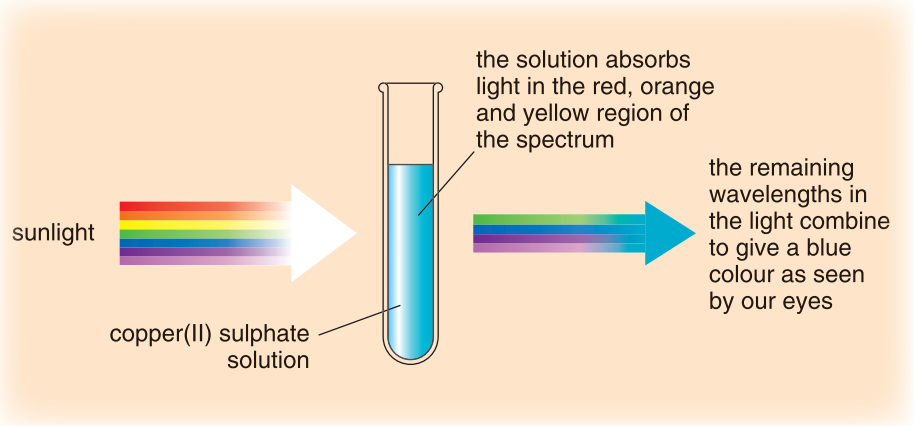


Fig. 42.6 What happens if you pass sunlight through copper(II) sulphate solution

The spectra in this diagram are for illustration only and are not real.

Table 42.3 lists the colours of aqueous ions of some transition metals.

Table 42.3

Colours of aqueous ions of some transition metals

Ion	Formula	Colour
Scandium(III)	$Sc^{3+}(aq)$	colourless
Titanium(III)	$Ti^{3+}(aq)$	purple
Vanadium(III)	$V^{3+}(aq)$	green
Chromium(III)	$Cr^{3+}(aq)$	green
Manganese(II)	$Mn^{2+}(aq)$	colourless (or pale pink)
Iron(II)	$Fe^{2+}(aq)$	pale green
Iron(III)	$Fe^{3+}(aq)$	yellow-brown
Cobalt(II)	$Co^{2+}(aq)$	pink
Nickel(II)	$Ni^{2+}(aq)$	green
Copper(II)	$Cu^{2+}(aq)$	blue
Zinc ion	$Zn^{2+}(aq)$	colourless
Permanganate	$MnO_4^-(aq)$	purple
Dichromate	$Cr_2O_7^{2-}(aq)$	orange

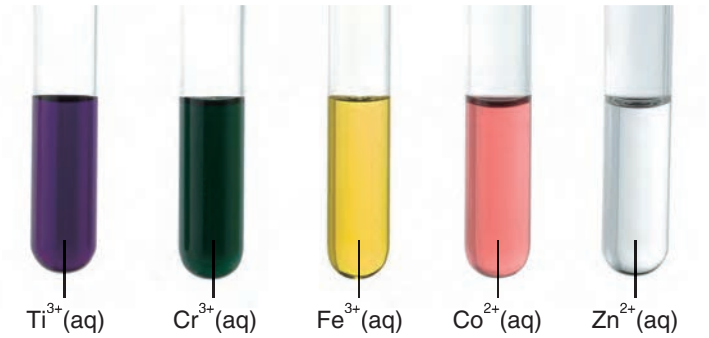


Fig. 42.7 Colours of aqueous ions of some transition metals

The majority of transition metal ions are coloured (Fig. 42.7). This aspect contrasts with that of the main group metals, whose ions are colourless. For example, sodium chloride, potassium nitrate and calcium oxide are all colourless.

Fig. 42.8 and Fig. 42.9 show how transition metal compounds can be used to add different colours in our daily life.



Fig. 42.8 Van Gogh used a chromium compound to provide his yellow colours

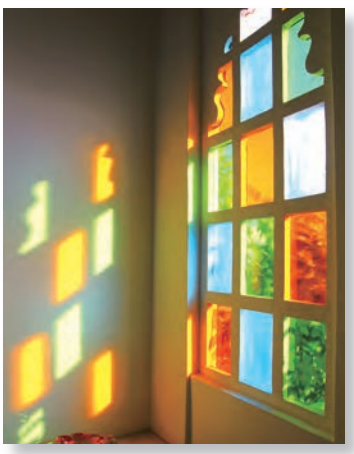


Fig. 42.9 Glass can be coloured by adding small amounts of transition metal compounds

42.3 Variable oxidation states of transition metals

The transition metals differ from most main group metals in that they show more than one oxidation state. For example, lithium, magnesium and aluminium have a single oxidation state equal to their group number.

Table 42.4 lists the formulae of the common oxides and chlorides of the elements scandium to zinc. All possible oxidation states of the elements are shown below these formulae.

Table 42.4

Formulae of the common oxides and chlorides of the elements scandium to zinc and all possible oxidation states of the elements

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Common oxide(s)	Sc_2O_3	Ti_2O_3 TiO_2	V_2O_3 V_2O_5	Cr_2O_3 CrO_3	MnO MnO_2 Mn_2O_7	FeO Fe_2O_3	CoO Co_2O_3	NiO	Cu_2O CuO	ZnO
Common chloride(s)	$ScCl_3$	$TiCl_3$ $TiCl_4$	VCl_3	$CrCl_2$ $CrCl_3$	$MnCl_2$ $MnCl_3$	$FeCl_2$ $FeCl_3$	$CoCl_2$	$NiCl_2$	CuCl $CuCl_2$	$ZnCl_2$
Possible oxidation state(s) of element in compounds					+7					
				+6	+6	+6				
			+5	+5	+5	+5	+5	+5		
		+4	+4	+4	+4	+4	+4	+4		
	+3	+3	+3	+3	+3	+3	+3	+3	+3	
		+2	+2	+2	+2	+2	+2	+2	+2	+2
		+1	+1	+1	+1	+1	+1	+1	+1	

The common oxidation states for each element are highlighted.

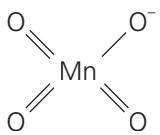
Notice that the number of possible oxidation states shown by an element increases from scandium to manganese. After manganese, there is a decrease in the number of possible oxidation states, reaching a minimum at zinc.

N2 Refer to page T18.

The general trends are as follows:

- Common oxidation states for each element include +2 or +3 or both. From titanium to chromium, +3 is the most common while from manganese onwards, +2 is the most common. N2
- The lower oxidation states are found in simple ions, such as Cr^{3+}, Mn^{2+}, Fe^{3+}, Cu^{2+}, etc.
- When the elements exhibit very high oxidation numbers (+4 or above), they do not form simple ions. Either they are involved in covalent bonding (as in Ti_2O_3, CrO_3 and Mn_2O_7) or they form polyatomic ions (such as CrO_4^{2-} and MnO_4^-).

The manganese atom in an MnO_4^- ion is covalently bonded to the four oxygen atoms by one single and three double bonds:



- Ions with the transition metals in a high oxidation state tend to be good oxidizing agents (e.g. $Cr_2O_7^{2-}$ and MnO_4^-).
- For transition metals at the beginning of the series, ions with the metals in a low oxidation state are good reducing agents (e.g. V^{2+} and Cr^{2+}).

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

42.2
Illustrating the oxidation states of vanadium.

42.4 Variable oxidation states of vanadium

Vanadium exhibits four common oxidation states, +2, +3, +4 and +5, of which +4 is considered to be the most stable under ordinary conditions. Vanadium(II) compounds are reducing agents. Vanadium(V) compounds are oxidizing agents. Thus, the chemistry of vanadium compounds typically involves a conversion from one oxidation state into another.

An acidic solution of a vanadium(V) compound can be prepared by dissolving ammonium metavanadate (NH_4VO_3) in sodium hydroxide solution and then adding dilute sulphuric acid. The solution has a yellow colour. Under these conditions, the main ions present in the solution are the **dioxovanadium(V) ions** ($VO_2^+(aq)$).

The exact vanadium ion present in the solution is very complicated, and varies with the pH of the solution.

$$VO_3^-(aq) + 2H^+(aq) \longrightarrow VO_2^+(aq) + H_2O(l)$$

The oxidation state of vanadium is +5 in both VO_3^- ion and VO_2^+ ion.

When the yellow solution is shaken with zinc powder, the dioxovanadium(V) ions change gradually to blue **oxovanadium(IV) ions** ($VO^{2+}(aq)$), then to green vanadium(III) ions ($V^{3+}(aq)$), and finally to violet vanadium(II) ions ($V^{2+}(aq)$) (Fig. 42.10).

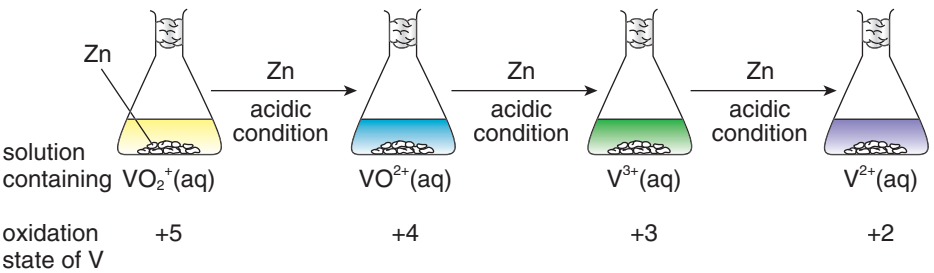


Fig. 42.10 Reduction that occurs when a yellow solution containing $VO_2^+(aq)$ ions is shaken with zinc powder under acidic conditions

If the reaction takes place in a small flask, it is normally stoppered with some cotton wool. This allows hydrogen (produced from a side reaction between zinc and the acid) to escape.

dioxovanadium(V) ion 二氧合釩(V)離子　　oxovanadium(IV) ion 氧合釩(IV)離子

Fig. 42.11 shows the aqueous solutions of various ions of vanadium while Table 42.5 lists their colours.

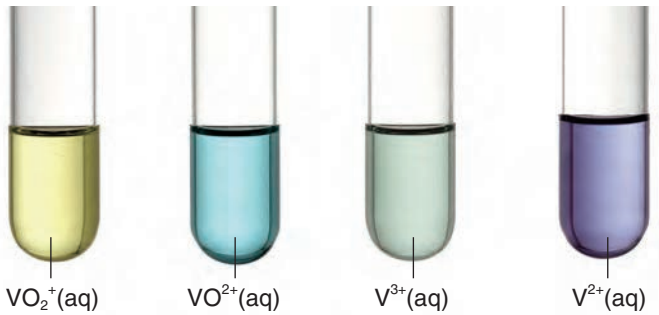


Fig. 42.11 Aqueous solutions of various ions of vanadium. Vanadium exists in different oxidation states in these ions (left to right: +5, +4, +3, +2)

Table 42.5

Colours of aqueous solutions of various vanadium ions

Oxidation state	Name of ion	Formula	Colour in aqueous solution
+5	dioxovanadium(V) ion	VO_2^+	yellow
+4	oxovanadium(IV) ion	VO^{2+}	blue
+3	vanadium(III) ion	V^{3+}	green
+2	vanadium(II) ion	V^{2+}	violet

Explaining reductions that occur when a solution containing dioxovanadium(V) ions is shaken with zinc powder N3

N3 Refer to page T19.

AL 2005 Paper 1 Q3(c)(i)

The following reductions occur when a solution containing dioxovanadium(V) ions ($VO_2^+(aq)$) is shaken with zinc powder under acidic conditions.

$$\overset{+5}{VO_2^+(aq)} \longrightarrow \overset{+4}{VO^{2+}(aq)} \longrightarrow \overset{+3}{V^{3+}(aq)} \longrightarrow \overset{+2}{V^{2+}(aq)}$$

Upon looking at the electrochemical series, we find the following ionic half-equations relevant to the above reductions (Fig. 42.12).

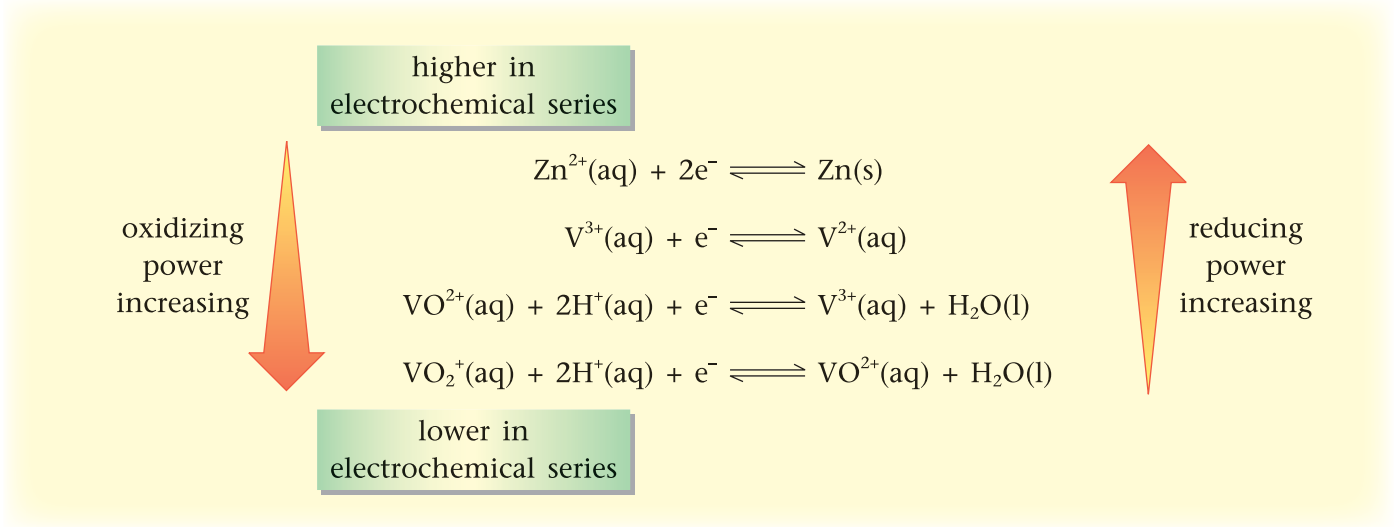
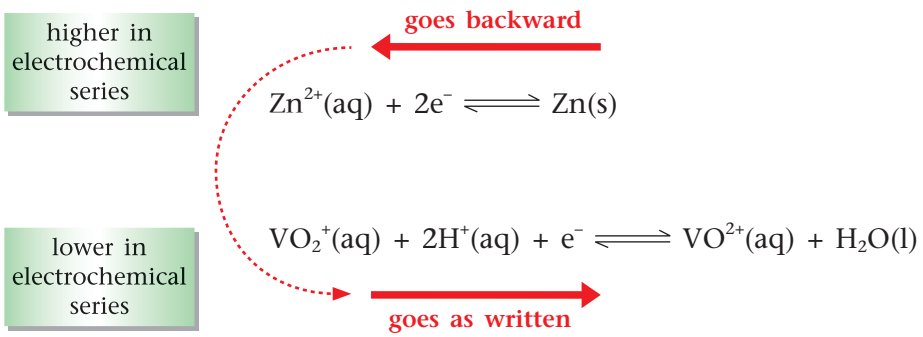


Fig. 42.12 Ionic half-equations in the electrochemical series relevant to the reduction of $VO_2^+(aq)$ ion by zinc powder

We have learnt in Topic 5 Redox Reactions, Chemical Cells and Electrolysis that when predicting the feasibility of a redox reaction, the process represented by the ionic half-equation higher in the series goes backward while the process represented by the ionic half-equation lower in the series goes as written (i.e. the anticlockwise rule).

Let us look at the first reduction — from $VO_2^+(aq)$ ion to $VO^{2+}(aq)$ ion. From the electrochemical series, we know that $VO_2^+(aq)$ ion is a stronger oxidizing agent than $Zn^{2+}(aq)$. Hence it will oxidize Zn(s) to $Zn^{2+}(aq)$ ion.

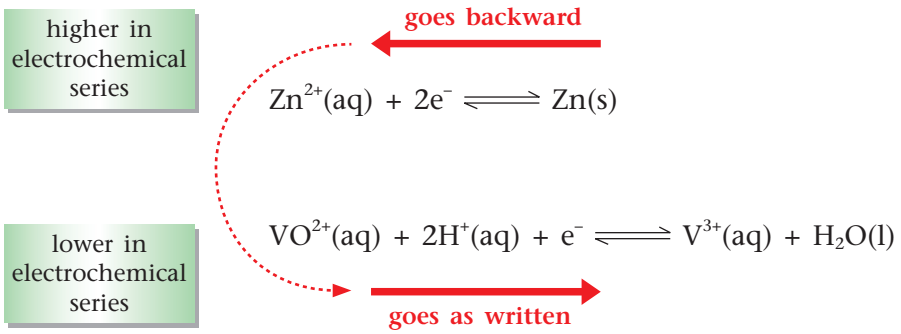


Combine the relevant ionic half-equations to obtain the ionic equation for the reduction from $VO_2^+(aq)$ ion to $VO^{2+}(aq)$ ion.

ASL 2012 Paper 1 Q2(c)
AL 2012 Paper 1 Q2(b)(iii)

$$2 \times (VO_2^+(aq) + 2H^+(aq) + e^- \longrightarrow VO^{2+}(aq) + H_2O(l))$$
$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$$
$$2VO_2^+(aq) + 4H^+(aq) + Zn(s) \longrightarrow 2VO^{2+}(aq) + 2H_2O(l) + Zn^{2+}(aq)$$

The second step is the reduction of the $VO^{2+}(aq)$ ion.

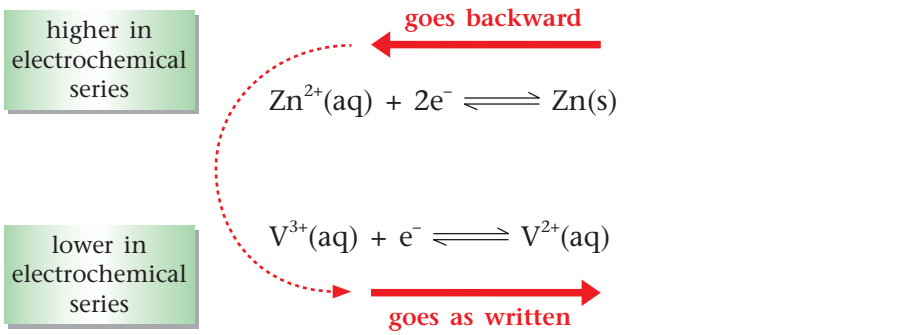


Combine the relevant ionic half-equations to obtain the ionic equation for the reduction from $VO^{2+}(aq)$ ion to $V^{3+}(aq)$ ion.

ASL 2012 Paper 1 Q2(c)
AL 2012 Paper 1 Q2(b)(iii)

$$2 \times (VO^{2+}(aq) + 2H^+(aq) + e^- \longrightarrow V^{3+}(aq) + H_2O(l))$$
$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$$
$$2VO^{2+}(aq) + 4H^+(aq) + Zn(s) \longrightarrow 2V^{3+}(aq) + 2H_2O(l) + Zn^{2+}(aq)$$

The third step is the reduction of the $V^{3+}(aq)$ ion.



Combine the relevant ionic half-equations to obtain the ionic equation for the reduction from $V^{3+}(aq)$ ion to $V^{2+}(aq)$ ion. N4

N4 Students should be able to write the ionic equation for the reaction between $VO_2^+(aq)$ ion and $Zn(s)$ in an acidic medium to give $V^{2+}(aq)$ ion and $Zn^{2+}(aq)$ ion.
$2VO_2^+(aq) + 3Zn(s) + 8H^+(aq) \longrightarrow 2V^{2+}(aq) + 3Zn^{2+}(aq) + 4H_2O(l)$

$$2 \times (V^{3+}(aq) + e^- \longrightarrow V^{2+}(aq))$$
$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$$
$$2V^{3+}(aq) + Zn(s) \longrightarrow 2V^{2+}(aq) + Zn^{2+}(aq)$$

ASL 2012 Paper 1 Q2(c)
AL 2012 Paper 1 Q2(b)(iii)

42.5 Variable oxidation states of chromium

Chromium exhibits three common oxidation states, +2, +3 and +6, of which +3 is the most stable.

Chromium reacts with acids in the absence of oxygen to give blue chromium(II) ions.

$$Cr(s) + 2H^+(aq) \longrightarrow Cr^{2+}(aq) + H_2(g)$$

In air, the reaction of chromium with acids yields green chromium(III) ions because chromium(II) ions are rapidly oxidized by atmospheric oxygen.

$$4Cr^{2+}(aq) + O_2(g) + 4H^+(aq) \longrightarrow 4Cr^{3+}(aq) + 2H_2O(l)$$

Chromium has its highest oxidation state of +6 in both the yellow chromate ion (CrO_4^{2-}) and the orange dichromate ion ($Cr_2O_7^{2-}$).

ASL 2010 Paper 1 Q5(a)
ASL 2011 Paper 2 Q4(a)
AL 2010 Paper 1 Q3(b)(i)
AL 2011 Paper 2 Q6(a)

These two species exist in an equilibrium. The position of equilibrium shifts to the right upon the addition of an acid and to the left upon the addition of an alkali. N5

$$\underset{\text{yellow}}{CrO_4^{2-}(aq)} + 2H^+(aq) \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + H_2O(l)$$

N5 Examination questions often ask about the conversion between $CrO_4^{2-}(aq)$ ion and $Cr_2O_7^{2-}(aq)$ ion.

Dichromate ion is a powerful oxidizing agent in acidic solutions. In the presence of a reducing agent, dichromate ion is reduced to chromium(III) ion.

$$\underset{\text{orange}}{\overset{+6}{Cr_2O_7^{2-}}(aq)} + 14H^+(aq) + 6e^- \longrightarrow \underset{\text{green}}{\overset{+3}{2Cr^{3+}}(aq)} + 7H_2O(l)$$

Fig. 42.13 shows the aqueous solutions of various ions of chromium.

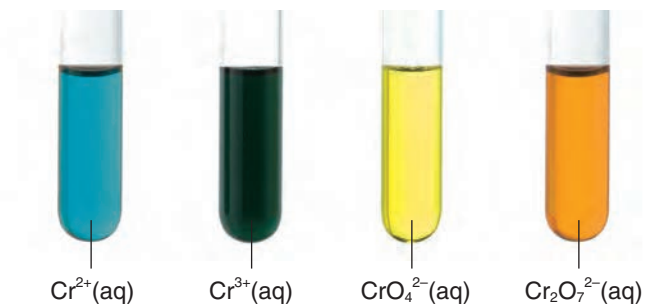


Fig. 42.13 Aqueous solutions of various ions of chromium. Chromium exists in different oxidation states in these ions (left to right: +2, +3, +6, +6)

42.6 Variable oxidation states of manganese

Manganese exhibits several common oxidation states, including +2, +4, +6 and +7. The +2 oxidation state is generally considered to be the most stable under normal conditions.

Manganese is easily raised to its +2 oxidation state, for example, by reaction with hydrochloric acid to form manganese(II) chloride ($MnCl_2$). The most common compound of manganese in its +4 oxidation state is manganese(IV) oxide (MnO_2).

Manganese has the +6 oxidation state in manganate ion (MnO_4^{2-}) and the +7 oxidation state in permanganate ion (MnO_4^-).

Fig. 42.14 shows the compounds or ions of manganese in different oxidation states while Table 42.6 lists their colours.

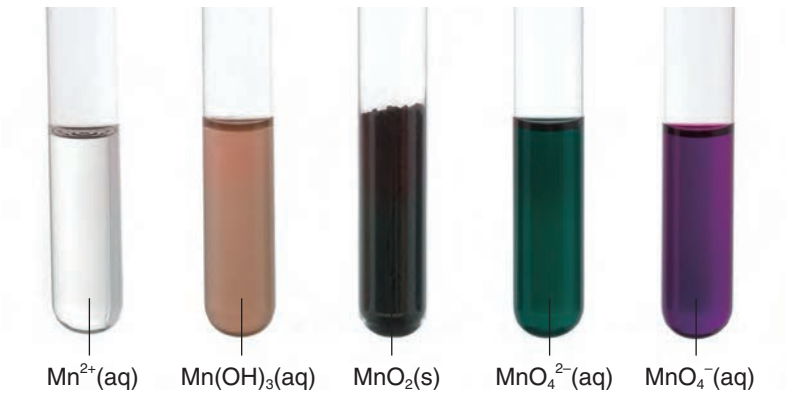


Fig. 42.14 Compounds or ions of manganese in different oxidation states (left to right: +2, +3, +4, +6, +7)

Table 42.6

Colours of compounds or ions of manganese in different oxidation states

Oxidation state	Compound / ion	Formula	Colour
+2	manganese(II) ion	Mn^{2+}	colourless (or pale pink)
+3	manganese(III) hydroxide	$Mn(OH)_3$	brown
+4	manganese(IV) oxide	MnO_2	black
+6	manganate ion	MnO_4^{2-}	dark green
+7	permanganate ion	MnO_4^-	purple

Permanganate ion is a powerful oxidizing agent. In the presence of dilute sulphuric acid and a reducing agent, permanganate ion is reduced to manganese(II) ion.

$$\overset{+7}{MnO_4^-}(aq) + 8H^+(aq) + 5e^- \longrightarrow \overset{+2}{Mn^{2+}}(aq) + 4H_2O(l)$$

purple → colourless (or pale pink)

However, under alkaline conditions, it reacts with a reducing agent to give manganese(IV) oxide.

$$\overset{+7}{MnO_4^-}(aq) + 2H_2O(l) + 3e^- \longrightarrow \overset{+4}{MnO_2}(s) + 4OH^-(aq)$$

purple → black

42.7 Variable oxidation states of iron

The most common oxidation states of iron are +2 and +3 (Fig. 42.15).



Fig. 42.15 Solutions containing the $Fe^{2+}(aq)$ ions (left) and the $Fe^{3+}(aq)$ ions (right)

When iron reacts with dilute hydrochloric acid in the absence of air, the product is a pale green solution of iron(II) ions.

$$Fe(s) + 2H^+(aq) \longrightarrow Fe^{2+}(aq) + H_2(g)$$

ASL 2009 Paper 2 Q6(c)
AL 2009 Paper 2 Q3(c)(iii)

In air, the iron(II) ions are slowly oxidized by atmospheric oxygen to the yellow-brown iron(III) ions.

$$4Fe^{2+}(aq) + O_2(g) + 4H^+(aq) \longrightarrow 4Fe^{3+}(aq) + 2H_2O(l)$$

When iron is treated with an acid having oxidizing property, such as dilute nitric acid, the metal is oxidized directly to iron(III) ions.

CE 2010 Paper 2 Q47

$$Fe(s) + NO_3^-(aq) + 4H^+(aq) \longrightarrow Fe^{3+}(aq) + NO(g) + 2H_2O(l)$$

42.3

Predicting the feasibility of reactions — redox reactions of iron.

Iron(II) salts in solution are common reducing agents. We can N6 represent the chemical change involved by the following ionic half-equation:

$$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$$

DSE 2012 Paper 1A Q30

N6 $Fe^{2+}(aq)$ ion can also act as an oxidizing agent.

Iron(III) salts in solution are common oxidizing agents. We can represent the chemical change involved by the following ionic half-equation:

$$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$$

Example 42.1

Q Consider the following ionic half-equations in the electrochemical series:

higher in electrochemical series

$$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$$

$$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$$

$$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$$

lower in electrochemical series

$$O_2(g) + 4H^+(aq) + 4e^- \rightleftharpoons 2H_2O(l)$$

$$Co^{3+}(aq) + e^- \rightleftharpoons Co^{2+}(aq)$$

What would happen if an aqueous solution containing the species in each of the following cases is exposed to air? Explain your answer in each case.

a) $V^{3+}(aq)$ ions

b) $Co^{3+}(aq)$ ions

AL 2006 Paper 1 Q1(c)

A a) $V^{3+}(aq)$ ions will be oxidized to $VO^{2+}(aq)$ ions.

$O_2(g)$ is a stronger oxidizing agent than $VO^{2+}(aq)$ ion.

The following reaction will occur:

$$4V^{3+}(aq) + O_2(g) + 2H_2O(l) \longrightarrow 4VO^{2+}(aq) + 4H^+(aq)$$

b) $Co^{3+}(aq)$ ions will be reduced to $Co^{2+}(aq)$ ions.

$Co^{3+}(aq)$ ion is a stronger oxidizing agent than $O_2(g)$.

The following reaction will occur:

$$4Co^{3+}(aq) + 2H_2O(l) \longrightarrow 4Co^{2+}(aq) + 4H^+(aq) + O_2(g)$$

Practice 42.1

Refer to page T19.

1 Consider the following ionic half-equations in the electrochemical series:

higher in electrochemical series

$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$

$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq)$

$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$

lower in electrochemical series

In an experiment, excess KI(aq) was added to a solution containing $VO_2^+(aq)$ ions. State the expected observations and write an ionic equation for the reaction involved.

2 'Scandium and zinc are not usually classified as transition metals.'

Justify the above statement by comparing the chemical properties of scandium and zinc with those of the transition metals.

42.8 Transition metals and their compounds as catalysts

We have discussed catalysts in Topic 10 Rate of Reaction and will discuss the topic again in Topic 13 Industrial Chemistry.

✓ A **catalyst** is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes.

Many transition metals and their compounds are effective catalysts, and are used in reactions both in the laboratory and in industry.

In Topic 10 Rate of Reaction, we have seen the decomposition of hydrogen peroxide to water and oxygen:

$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

At room temperature, this reaction is very slow. However, if manganese(IV) oxide is added, it acts as a catalyst and the reaction becomes very rapid.

Nowadays, about 80% of processes in the chemical industry depend on catalysts to work efficiently.

catalyst 催化劑

Do you know

Autocatalysis

An interesting example of catalysis occurs when one of the products of a reaction is a catalyst for the reaction — autocatalysis.

One example is the reaction between potassium permanganate and sodium ethanedioate under acidic conditions. This reaction is catalyzed by $Mn^{2+}(aq)$ ions.

$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$

The catalyst $Mn^{2+}(aq)$ ions are not present at the beginning of the reaction. Once $Mn^{2+}(aq)$ ions are formed, they can react with $MnO_4^-(aq)$ ions to form $Mn^{3+}(aq)$ ions:

$4Mn^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \longrightarrow 5Mn^{3+}(aq) + 4H_2O(l)$

The $Mn^{3+}(aq)$ ions can then react with $C_2O_4^{2-}(aq)$ ions to liberate $CO_2(g)$ and reform $Mn^{2+}(aq)$ ions, which can then continue the autocatalysis.

$2Mn^{3+}(aq) + C_2O_4^{2-}(aq) \longrightarrow 2CO_2(g) + 2Mn^{2+}(aq)$

Haber process for the manufacture of ammonia

In the **Haber process**, nitrogen and hydrogen react to give ammonia. Finely divided iron is used as a catalyst in the reaction.

$$N_2(g) + 3H_2(g) \rightleftharpoons^{\text{Fe; 400 °C–450 °C, 200 atmospheres}} 2NH_3(g)$$

Contact process for the manufacture of sulphuric acid

During the manufacture of sulphuric acid in the **Contact process**, sulphur dioxide is converted into sulphur trioxide by passing sulphur dioxide and oxygen over a solid vanadium(V) oxide catalyst.

CE 2006 Paper 2 Q20
CE 2010 Paper 2 Q15

$$2SO_2(g) + O_2(g) \rightleftharpoons^{V_2O_5;\ 450\ °C–600\ °C,\ 1\ \text{atmosphere}} 2SO_3(g)$$

Fig. 42.16 shows a Contact process plant used for making sulphuric acid.



Fig. 42.16 A Contact process plant used for making sulphuric acid

Haber process 哈柏法　　Contact process 接觸法

The catalytic action of vanadium(V) oxide takes place in two steps as follows:

- The vanadium(V) oxide oxidizes sulphur dioxide to sulphur trioxide and is itself reduced to vanadium(IV) oxide:

$$\overset{+5}{V_2O_5}(s) + SO_2(g) \longrightarrow \overset{+4}{V_2O_4}(s) + SO_3(g)$$

- The vanadium(IV) oxide is then oxidized back to vanadium(V) oxide by oxygen:

$$2\overset{+4}{V_2O_4}(s) + O_2(g) \longrightarrow 2\overset{+5}{V_2O_5}(s)$$

The vanadium(V) oxide is regenerated unchanged. This is a good example of how the variability of oxidation states of a transition metal is useful in catalysis.

Manufacture of nitric acid from ammonia

We will discuss this manufacturing process in Topic 13 Industrial Chemistry.

The first stage in the manufacture of nitric acid from ammonia involves the catalytic oxidation of ammonia to nitrogen monoxide. Ammonia is oxidized by reacting excess air and ammonia at 900 °C and about 10 atmospheres, using a platinum-rhodium catalyst (Fig. 42.17).

$$4NH_3(g) + 5O_2(g) \underset{900\ °C,\ 10\ \text{atmospheres}}{\overset{\text{Pt-Rh}}{\rightleftharpoons}} 4NO(g) + 6H_2O(g)$$



Fig. 42.17 During the production of nitric acid, gauze of platinum and rhodium provides a huge surface area for the catalytic oxidation of ammonia

Table 42.7 lists examples of reactions in which transition metals and their compounds act as catalysts.

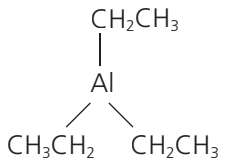
N7 Refer to page T20.

Table 42.7

Examples of reactions in which transition metals and their compounds act as catalysts N7

Catalyst	Reaction catalyzed
Iron (Fe)	Haber process $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Vanadium(V) oxide (V_2O_5)	Contact process $2SO_2 + O_2 \rightleftharpoons 2SO_3$
Platinum-rhodium (Pt-Rh)	Manufacture of nitric acid from ammonia $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$
Triethylaluminium◂ and titanium(IV) chloride ($(C_2H_5)_3Al$ / $TiCl_4$)	Polymerization of ethene to form high density polythene $n\,CH_2{=}CH_2 \longrightarrow \left[CH_2{-}CH_2\right]_n$
Nickel (Ni)	Hydrogenation of unsaturated vegetable oils (production of margarines) $-CH{=}CH- + H_2 \longrightarrow -CH_2{-}CH_2-$
Copper or copper(II) oxide (Cu or CuO)	Oxidation of ethanol to ethanal $CH_3CH_2OH + \frac{1}{2}O_2 \longrightarrow CH_3CHO + H_2O$

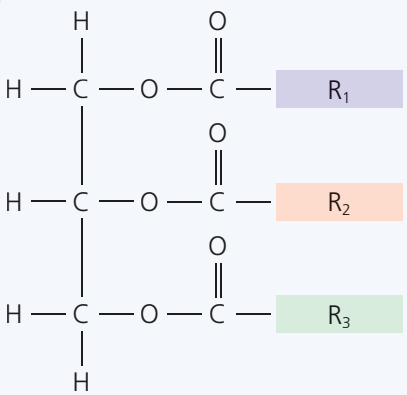
◂ Triethylaluminium consists of ethyl groups covalently bonded to aluminium atoms.



Do you know

How margarines are produced?

Fats and oils are mixed triglycerides. A triglyceride is formed when glycerol (i.e. propane-1,2,3-triol) combines with three carboxylic acids having long hydrocarbon chains.



a triglyceride

where R_1, R_2 and R_3 represent long hydrocarbon chains

A saturated carboxylic acid has only single bonds in its hydrocarbon chain, while an unsaturated carboxylic acid contains at least one C=C bond in its chain.

Triglycerides formed from largely saturated carboxylic acids are solids at room temperature. Triglycerides formed from a high proportion of unsaturated carboxylic acids are liquids at room temperature.

Vegetable oils contain triglycerides formed from a high proportion of unsaturated carboxylic acids. Thus, the hydrocarbon chains of these triglycerides contain many C=C bonds. When hydrogen is passed through a vegetable oil in the presence of a nickel catalyst, hydrogen adds to the C=C bonds and turns the oil into a solid fat. The process is called hydrogenation. *Margarines* are produced in this way.

margarine 人造牛油

42.9 Uses of transition metals in medicine

A tracer is a radioactive isotope whose presence in a system can easily be detected.

Radioactive isotopes of transition metals are widely used in medicine. For example, one radioactive isotope of chromium, chromium-51, is used as a *tracer* in studies on blood. The isotope is injected into the blood circulation system. Inside the system, the isotope gives off radiation. That radiation can be followed by detectors.

One radioactive isotope of cobalt, cobalt-60, is often used as a source of radiation in the treatment of *tumors* (Fig. 42.18).



Fig. 42.18 Cobalt-60 is used for cancer treatment

Two radioactive isotopes of copper are also used in medicine. One is copper-64. This isotope is used to study brain function and to detect *Wilson's disease*. This disease is the inability to eliminate copper from one's body. The second isotope is copper-67. This isotope can be used to treat cancer.

radioactive isotope 放射性同位素　tracer 示蹤物　tumor 腫瘤　Wilson's disease 威爾森氏症



Fig. 42.19 A bottle of tablets containing vitamin B_{12}

Importance of transition metals in maintaining health

Many transition metals are required by human beings and other living things in minute but definite amounts. Elements required in such small amounts are called **trace elements**. These trace elements include copper, manganese, iron, cobalt, nickel and chromium.

Cobalt occurs in animals in the form of vitamin B_{12}. The vitamin is necessary for building proteins for muscles and red blood cells. Human beings take in sufficient amounts of vitamin B_{12} from a normal diet. When absorption does not occur normally, an individual may have to take a vitamin B_{12} supplement (Fig. 42.19).

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Transition metal complex in medicine

A complex is formed when a transition metal ion is surrounded by other molecules or anions (called *ligands*) which form dative covalent bonds to the central metal ion.

Cis-platin (Fig. 42.20) is one of the most effective drugs against many forms of cancer. It is the *cis* isomer of a platinum complex, $[Pt(NH_3)_2Cl_2]$.

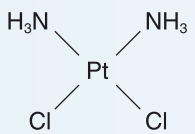


Fig. 42.20 Structure of *cis*-platin

Cells divide to make more cells of the same type. Before cell division, a copy is made of the DNA present in the cell. After cell division, two new cells are formed, each containing the same DNA.

Cancers form when some cells in the body multiply uncontrollably and abnormally. *Cis*-platin damages the DNA inside the cancer cells and so prevents them from multiplying.

However, *cis*-platin can produce some severe side effects. It is now mainly used in combination with other drugs — given as a cocktail.

Questions

1 a) What is the difference between a covalent bond and a dative covalent bond?

b) Both ammonia molecule and chloride ion can act as ligands.

What structural feature must a ligand have?

Has at least one pair of electrons.

2 The tetrachloroplatinate(II) ion readily undergoes the following reaction:

$$[PtCl_4]^x + 2NH_3 \rightleftharpoons [Pt(NH_3)_2Cl_2]^y + 2Cl^-$$

a) i) Suggest values for x and y in the equation. $x = -2$ $y = 0$

ii) The complex $[Pt(NH_3)_2Cl_2]^y$ exists in two isomeric forms. Draw the structure of the *trans* isomer.

b) The *cis* isomer is an important drug used in cancer treatment.

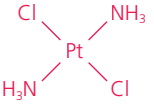
How does this drug help in cancer treatment?

It damages the DNA inside the cancer cells and prevents cell replication.

trace element 微量元素　　ligand 配位體

1 a) Both involve the sharing of a pair of electrons. In a covalent bond, each bonded atom supplies one electron to the bond. In a dative covalent bond, both electrons come from one atom.

2 a) ii)



Have you mastered?

Key terms

transition metal 過渡金屬 39 oxidation state 氧化態 43 electromagnetic radiation 電磁輻射 43

dioxovanadium(V) ion 二氧合釩(V)離子 47 oxovanadium(IV) ion 氧合釩(IV)離子 47

catalyst 催化劑 55 Haber process 哈柏法 56 Contact process 接觸法 56

radioactive isotope 放射性同位素 59 trace element 微量元素 60

Checklist

After studying this unit, you should be able to

- ☐ identify the positions of transition metals in the periodic table;
- ☐ recognize that most aqueous ions of transition metals are coloured;
- ☐ decribe the colours of common transition metal ions;
- ☐ describe that transition metals can exist in more than one oxidation states in their compounds;
- ☐ describe the reduction of dioxovanadium(V) ion by zinc and the ions produced;
- ☐ describe that transition metals and their compounds can be used as catalysts;
- ☐ describe the importance of transition metals.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Typical chemical properties of transition metals include:

- formation of coloured compounds;
- existence in more than one oxidation state in their compounds; and
- catalytic properties of the elements and their compounds.

2 The following table lists the colours of some ions of transition metals in aqueous solution:

Ion	Formula	Colour
Chromium(III)	$Cr^{3+}(aq)$	green
Manganese(II)	$Mn^{2+}(aq)$	colourless (or pale pink)
Iron(II)	$Fe^{2+}(aq)$	pale green
Iron(III)	$Fe^{3+}(aq)$	yellow-brown
Nickel(II)	$Ni^{2+}(aq)$	green
Copper(II)	$Cu^{2+}(aq)$	blue

3 The following reductions occur when a solution containing dioxovanadium(V) ions ($VO_2^+(aq)$) is shaken with zinc powder under acidic conditions.

	$\overset{+5}{VO_2^+(aq)}$	$\longrightarrow$	$\overset{+4}{VO^{2+}(aq)}$	$\longrightarrow$	$\overset{+3}{V^{3+}(aq)}$	$\longrightarrow$	$\overset{+2}{V^{2+}(aq)}$
Name	dioxovanadium(V) ions		oxovanadium(IV) ions		vanadium(III) ions		vanadium(II) ions
Colour	yellow		blue		green		violet

4 Dichromate ion is a powerful oxidizing agent in acidic solution. In the presence of a reducing agent, dichromate ion is reduced to chromium(III) ion.

$$\overset{+6}{Cr_2O_7^{2-}(aq)} + 14H^+(aq) + 6e^- \longrightarrow \overset{+3}{2Cr^{3+}(aq)} + 7H_2O(l)$$

orange → green

5 Permanganate ion is a powerful oxidizing agent. In the presence of dilute sulphuric acid and a reducing agent, permanganate ion is reduced to manganese(II) ion.

$$\overset{+7}{MnO_4^-(aq)} + 8H^+(aq) + 5e^- \longrightarrow \overset{+2}{Mn^{2+}(aq)} + 4H_2O(l)$$

purple → colourless (or pale pink)

6 The following table lists examples in which transition metals and their compounds act as catalysts.

Catalyst	Reaction catalyzed
Iron (Fe)	Haber process $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Vanadium(V) oxide (V_2O_5)	Contact process $2SO_2 + O_2 \rightleftharpoons 2SO_3$
Platinum-rhodium (Pt-Rh)	Manufacture of nitric acid from ammonia $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$

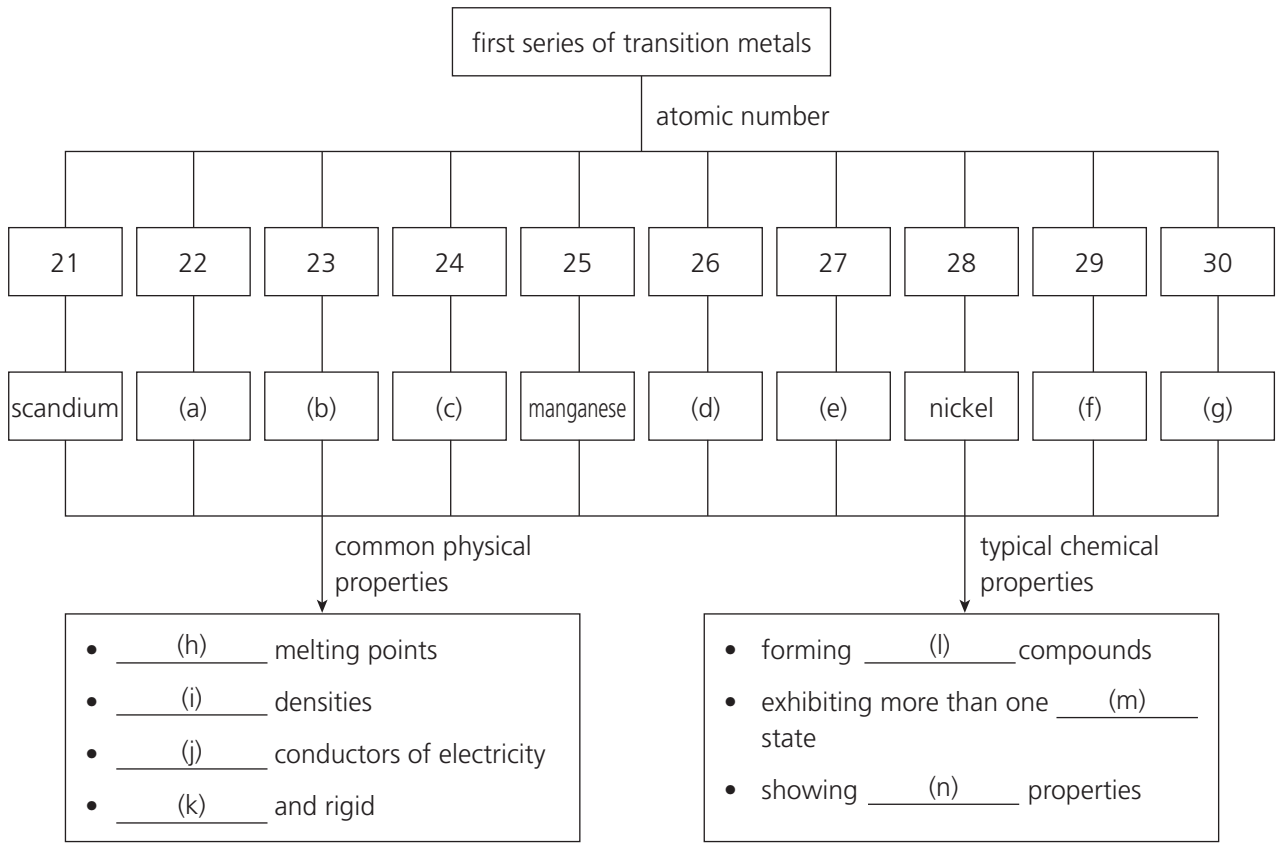
Have you mastered?

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) titanium
b) vanadium
c) chromium
d) iron
e) cobalt
f) copper
g) zinc
h) high
i) high
j) good
k) hard
l) coloured
m) oxidation
n) catalytic

2 Complete the table.

Ion	Colour
$Ti^{3+}(aq)$	purple
$V^{3+}(aq)$	green
$Cr^{3+}(aq)$	green
$Mn^{2+}(aq)$	colourless / pale pink
$Co^{2+}(aq)$	pink
$Ni^{2+}(aq)$	green
$MnO_4^-(aq)$	purple
$Cr_2O_7^{2-}(aq)$	orange

3 Give the oxidation state of the transition metal in each of the following species:

a) TiO_2 +4 b) VO_2^+ +5 c) VO^{2+} +4

d) VO_3^- +5 e) MnO_4^- +7 f) $Cr_2O_7^{2-}$ +6

Part II Multiple choice questions

4 Which of the following ions is responsible for the green colour of jade?

(A) Cr^{3+}
B Fe^{3+}
C Mn^{2+}
D Mn^{3+}

Option B — Fe^{3+} ions are yellow-brown in colour.
Option C — Mn^{2+} ions are colourless / pale pink in colour.

5 Which of the following combinations is correct?

	Species	Colour
A	$Mn^{2+}(aq)$	brown
B	$Mn(OH)_3(s)$	pale pink
C	$MnO_2(s)$	purple
(D)	$MnO_4^{2-}(aq)$	dark green

6 Which of the following species does NOT exist?

A $Sc^{3+}(aq)$
B $Mn_2O_7(l)$
C $TiCl_3(s)$
(D) $V^{5+}(aq)$

When elements exhibit very high oxidation numbers (+4 or above), they do not form simple ions.

7 Which of the following transition metals has a maximum oxidation state of +4?

A Co
B Mn
(C) Ti
D V

Option A — The maximum oxidation state of Co is +5.
Option B — The maximum oxidation state of Mn is +7.
Option D — The maximum oxidation state of V is +5.

8 Which of the following transition metals forms an oxide which is used in the manufacture of sulphuric acid?

A Chromium
B Manganese
(C) Vanadium
D Zinc

Vanadium(V) oxide is used as a catalyst in the manufacture of sulphuric acid.

9 The table below shows the possible oxidation states of four transition metals.

Transition metal	Possible oxidation states
W	+1 +2 +3 +4
X	+1 +2 +3 +4 +5
Y	+1 +2 +3 +4 +5 +6
Z	+1 +2 +3 +4 +5 +6 +7

Which of the following ions is likely to exist?

A WO_2^+
B XO_4^{2-}
C YO_4^-
(D) ZO_2^{2+}

The oxidation state of Z in ZO_2^{2+} is +6.
A possible oxidation state of Z is +6.
Thus, the ion is likely to exist.

Directions: Questions 10 and 11 refer to the following reaction scheme.

$VO_2^+(aq)$ —Cu(s), *Reaction 1*→ $VO^{2+}(aq)$ —Cu(s), *Reaction 2*→ $V^{3+}(aq)$

10 What is the colour of the final solution?

A Brownish black
(B) Greenish blue
C Reddish violet
D Yellowish brown

11 Which of the following statements are correct?
Refer to page T20 for explanation.

(1) Cu(s) acts as a reducing agent in the reaction.
(2) $VO_2^+(aq)$ ion is a stronger oxidizing agent than $Cu^{2+}(aq)$ ion.
(3) The oxidation states of vanadium shown in the reaction scheme are +6, +4 and +3.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

12 Which of the following statements about iron is / are correct?

(1) Its atom has an electronic arrangement of 2,8,14,1.
(2) It has a high density.
(3) Its common oxidation states are +1 and +2.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(1) An iron atom has an electronic arrangement of 2,8,14,2.
(3) The common oxidation states of iron are +2 and +3.

13 Which of the following ions is / are blue in colour?

(1) $Cu^{2+}(aq)$
(2) $Ti^{3+}(aq)$
(3) $VO_2^+(aq)$

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

14 The atom of element X has an electronic arrangement of 2,8,13,1. Which of the following statements concerning X are correct?

(1) The highest oxidation state X can exhibit is +7.
(2) X is found in stainless steel.
(3) X forms an oxide with a chemical formula X_2O_3.

A (1) and (2) only
B (1) and (3) only
(C) (2) and (3) only
D (1), (2) and (3)

Element X is chromium.
(1) The highest oxidation state X can exhibit is +6.

Part III Structured questions

15 State THREE characteristic chemical properties of transition elements. In each case, illustrate your answer with an example involving vanadium.

16 a) The common oxidation states of vanadium are shown in the table below. Complete the table.

	VO_2^+	VO^{2+}	V^{3+}	V^{2+}
Oxidation number of vanadium	+5	+4	+3	+2
Colour	yellow	blue	green	violet

b) Consider the following ionic half-equations in the electrochemical series: Refer to page T21.

higher in electrochemical series

$$V^{2+}(aq) + 2e^- \rightleftharpoons V(s)$$
$$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$$
$$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$$
$$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$$
$$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$$

lower in electrochemical series

Different oxidation states of vanadium may be observed by carrying out the successive reduction of ammonium metavanadate using zinc in an acidic solution.

i) Write an ionic equation for each of the following reduction process:

(1) $VO_2^+(aq)$ to $VO^{2+}(aq)$;
(2) $VO^{2+}(aq)$ to $V^{3+}(aq)$;
(3) $V^{3+}(aq)$ to $V^{2+}(aq)$.

ii) Explain why the reduction process stops at the $V^{2+}(aq)$ ion.

15 • They demonstrate catalytic activity.
e.g. vanadium(V) oxide is used as a catalyst in the Contact process.
• They exhibit variable oxidation states in their compounds.
e.g. the oxidation state of vanadium in VO_2^+ ion is +5, the oxidation state of vanadium in V^{3+} ion is +3.
• They form coloured compounds.
e.g. $V^{3+}(aq)$ ion is green in colour.

17 The following ionic half-equations are arranged in the order as shown in the electrochemical series: Refer to page T22.

higher in electrochemical series

$$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$$
$$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$$
$$O_2(g) + 4H^+(aq) + 4e^- \rightleftharpoons 2H_2O(l)$$

lower in electrochemical series

The table below lists the colours of three vanadium-containing species.

Species	Colour
$VO^{2+}(aq)$	blue
$V^{3+}(aq)$	green
$V^{2+}(aq)$	violet

An aqueous solution containing $V^{3+}(aq)$ ions was exposed to air.

a) State the expected observation and explain your answer.

b) Write an ionic equation for the reaction involved.

18 Acidified dichromate solution is a common oxidizing agent. The dichromate ion can be reduced to chromium(III) ion.

a) Write an ionic half-equation for the reduction of dichromate ion to chromium(III) ion in acidic solution.
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightleftharpoons 2Cr^{3+}(aq) + 7H_2O(l)$

b) State the colour change which occurs during this reduction process.
From orange to green

c) A reaction occurs when hydroxide ions are added to a solution containing dichromate ions.

i) State the colour change which occurs.
From orange to yellow

ii) Write an equation for the reaction.
$Cr_2O_7^{2-}(aq) + 2OH^-(aq) \rightleftharpoons 2CrO_4^{2-}(aq) + H_2O(l)$

iii) Explain whether this reaction is a redox reaction.
It is not a redox reaction.
The oxidation states of all the elements remain unchanged.

19 The water from wells in some rural areas smells of rotten eggs due to the presence of hydrogen sulphide from decaying vegetable and animal matter. The smell can be removed by the addition of dilute potassium permanganate solution.

a) i) Balance the equation below for the reaction removing the 'rotten egg' smell.

$$\underline{3}\ H_2S + \underline{2}\ MnO_4^- + 2H^+ \longrightarrow \underline{2}\ MnO_2 + \underline{3}\ S + 4H_2O$$

ii) Give the oxidation state of Mn in MnO_2. +4

b) The potassium permanganate will also remove any iron(II) ions, which are often present in water from wells, converting them to iron(III) ions.

A student added some drops of potassium permanganate solution to a sample of water from a well containing iron(II) ions. He then added excess sodium hydroxide solution and observed a red-brown precipitate.

i) Name the precipitate.
Iron(III) hydroxide

ii) Give the ionic equation for the formation of the precipitate. Include state symbols.
$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$

(OCR Advanced GCE, Chem. B (Salters), F334, Jan. 2010, 5(a)–(b))

20 A student plans to convert chloride ions into chlorine.

Consider the following ionic half-equations in the electrochemical series:

higher in electrochemical series

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightleftharpoons 2Cr^{3+}(aq) + 7H_2O(l)$$
$$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq)$$
$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$$

lower in electrochemical series

Suggest, with reasons, whether acidified potassium permanganate solution and acidified potassium dichromate solution would be suitable to convert chloride ions to chlorine.

Acidified potassium permanganate solution is suitable.

$MnO_4^-(aq)$ ion is a stronger oxidizing agent than $Cl_2(g)$. However, $Cr_2O_7^{2-}(aq)$ ion is a weaker oxidizing agent than $Cl_2(g)$.

21 Iron is the second most abundant metal in the Earth's crust. The oxidation states of iron in iron(II) sulphate and iron(III) sulphate are +2 and +3 respectively.
Refer to page T22.

a) Topaz, a yellow-coloured gemstone, is iron-containing. Suggest the ion of iron in topaz that is responsible for the yellow colour.

b) What will be observed when dilute sodium hydroxide solution is added to iron(II) sulphate solution and iron(III) sulphate solution respectively? Write ionic equations for the reactions involved.

c) When iron(II) sulphate solution was mixed with an acidified solution of orange colour, a reaction occurred and the oxidation state of iron changed from +2 to +3.

i) Suggest what the orange solution might be.

ii) What would be observed in this reaction? Write an ionic equation for the reaction involved.

22 Transition metals and their compounds are important industrial catalysts.

a) For each of the following metals, identify an industrial process which uses it or one of its compounds as a catalyst.

i) Iron

ii) Nickel

iii) Vanadium

b) For each process suggested in (a), write down an equation for the catalyzed reaction.

c) What property of transition metals is likely to be the most important in enabling them, or their compounds, to act as catalysts?

22 a) i) Iron as a catalyst in the Haber process.
ii) Nickel as a catalyst in the hydrogenation of unsaturated vegetable oils.
iii) Vanadium(V) oxide as a catalyst in the Contact process.

b) i) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
ii) $—CH=CH— + H_2 \longrightarrow —CH_2—CH_2—$
iii) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

c) Having variable oxidation states

Topic Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Multiple choice questions

1 In which of the following series does the melting point of the element increase from left to right?

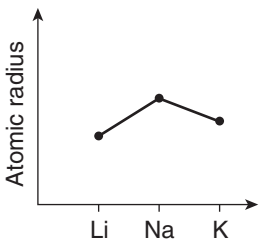
A Li, Na, K
B Al, Si, P
C Si, P, S
(D) Na, Mg, Al

The strength of metallic bonds increases in the order: Na, Mg, Al. Thus, the melting point increases in the same order.

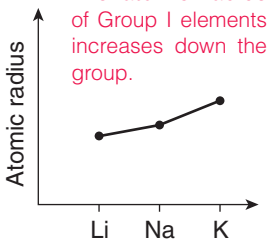
(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 16)

2 Which of the following graphs shows the variation of atomic radius of Group I elements?

A

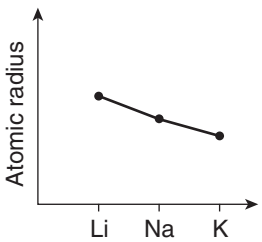


(B)

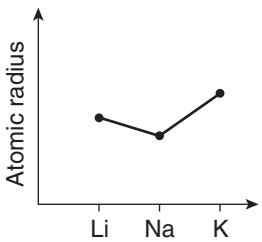


The atomic radius of Group I elements increases down the group.

C



D



3 Which of the following sets contains two covalent oxides and two ionic oxides?

A Al_2O_3, Br_2O_5, CO, SO_3
B B_2O_3, CO_2, NO_2, Na_2O
C BaO, Cl_2O_7, F_2O, SiO_2
(D) CaO, Cl_2O, K_2O, P_4O_6

CaO and K_2O are ionic oxides while Cl_2O and P_4O_6 are covalent oxides.

4 What is the oxidation state of iron in $BaFeO_4$?

A +2
B +4
(C) +6
D +8

5 Which of the following oxides would form an acidic solution when added to water?

(A) Carbon dioxide
B Silicon dioxide
C Aluminium oxide
D Lithium oxide

(HKDSE, Paper 1A, 2012, 31)

6 Which of the following gives the colours of solutions containing $Cr_2O_7^{2-}$, CrO_4^{2-}, hydrated Cr^{3+} and hydrated Cr^{2+}, in this order?

Refer to the next page for explanation.

A Yellow, orange, green, blue
(B) Orange, yellow, green, blue
C Orange, yellow, blue, green
D Orange, green, yellow, blue

(Edexcel Advanced GCE, Unit 5, Jun. 2010, 8)

7 Which of the following ions can act as both an oxidizing agent and a reducing agent?

(A) $Fe^{2+}(aq)$
B $Cu^{2+}(aq)$
C $Cr_2O_7^{2-}(aq)$
D $MnO_4^-(aq)$

(HKDSE, Paper 1A, 2012, 30)

8 Which of the following is NOT a characteristic property of transition metals?

A They form coloured compounds.
B They exhibit variable oxidation numbers in their compounds.
C They react with dilute hydrochloric acid to give hydrogen gas.
D They exhibit catalytic property in elemental state or as compounds.

(HKDSE, Paper 1A, 2013, 26)

9 Comparing the elements in the second period of the periodic table, from lithium to fluorine, which of the following statements is / are correct?

(1) They show a gradual change from having metallic property to having non-metallic property.
(2) They show a gradual increase in the number of electron shells in their atoms.
(3) They show a gradual decrease in melting point.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 22)

10 There is a gradual change in the properties of the three halogens, chlorine, bromine and iodine. Which of the following properties is / are in the order of chlorine > bromine > iodine?

(1) Atomic radius
(2) Electronegativity
(3) Oxidizing power

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) Atomic radius of the halogens increases in the order: chlorine < bromine < iodine.
(2) Electronegativity of the halogens increases in the order: chlorine > bromine > iodine.
(3) Oxidizing power of the halogens increases in the order: chlorine > bromine > iodine.

11 Which of the following is / are amphoteric oxide(s)?

(1) Al_2O_3
(2) B_2O_3
(3) Ga_2O_3

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) B_2O_3 is an acidic oxide.

12 In which of the following reactions, is / are the transition metal species NOT acting as a catalyst?

(1) Action of acidified MnO_4^-(aq) on SO_3^{2-}(aq) at room temperature
(2) Action of Ni(s) on a mixture of $H_2C{=}CH_2$(g) and H_2(g) at high temperature
(3) Action of Pt(s) on a mixture of CO(g) and O_2(g) at high temperature

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 30)

6

Solution containing	Colour
$Cr_2O_7^{2-}$	orange
CrO_4^{2-}	yellow
hydrated Cr^{3+}	green
hydrated Cr^{2+}	blue

Directions : *Each question (Questions 13–14) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
13	The melting point of the non-metals in Period 3 of the periodic table decreases from sulphur to argon.	The relative atomic mass increases from sulphur to argon in Period 3 of the Periodic Table.	B
		(HKDSE, Practice paper 1A, 2012, 35)	
14	Both aluminium oxide and magnesium oxide exhibit similar acid-base properties.	Both aluminium oxide and magnesium oxide are ionic oxides.	C
		(HKDSE, Paper 1A, 2013, 36)	

Topic Exercise

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

15 State and explain the trend in atomic radius of the Period 3 elements from sodium to chlorine.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2009, 1(c))

16 Silicon exists in a giant covalent lattice.

a) The electrical conductivity of pure silicon is very low.

Explain why this is so in terms of the type of bonding.

b) Explain the high melting point of silicon in terms of the type of bonding.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 26(c))

17 Sketch the variation of the melting points of the following elements: Na, Mg, Si, S and Cl. Account for the variation.

(HKASLE, Paper 2, 2012, 5)

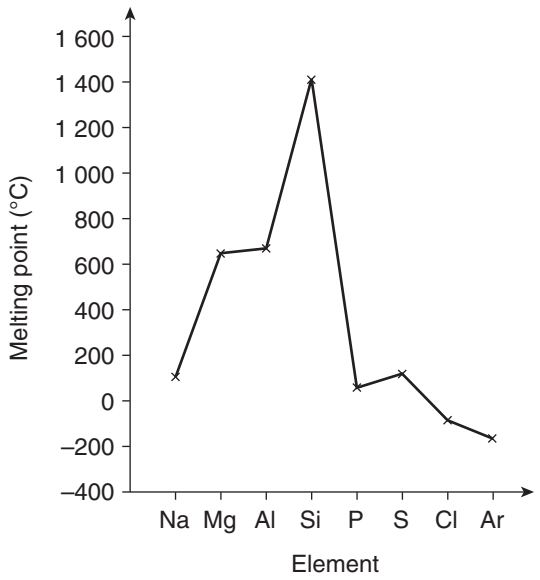
18 Using potassium and iron as examples, suggest TWO similarities and TWO differences between the properties of alkali metals and transition metals.
Refer to page T24.

19 Sketch the variation of the electrical conductivity of the following elements: Na, Mg and Al. Account for the variation.
Refer to page T24.

15 Decrease
Moving across Period 3 from sodium to chlorine, the number of protons increases by one with each successive element.
The shielding effect provided by the two inner electron shells remain roughly the same for each element.
So, the greater attractive force exerted by the increasing positive nuclear charge on the outermost shell electrons pull these electrons closer to the nucleus.

16 a) In silicon, the four outermost shell electrons in each atom are held in four covalent bonds.
At room temperature, very few electrons gain enough energy to become delocalized.
b) Strong covalent bonds exist between silicon atoms.
To melt silicon, many of the covalent bonds between the atoms must be broken. This needs a lot of heat.

20 The graph shows the variation in melting points of the elements across Period 3 (Na to Ar) of the periodic table.



a) Complete the table below to show the type of structure and bonding for the elements shown.

Element	Structure	Bonding
Sodium	giant metallic	metallic
Silicon	giant covalent	covalent
Sulphur	simple molecular	covalent bond holding atoms together within the molecule; van der Waals' forces between molecules

b) Explain why silicon has a much higher melting point than sulphur.

c) Explain why the melting point increases from sodium to aluminium.

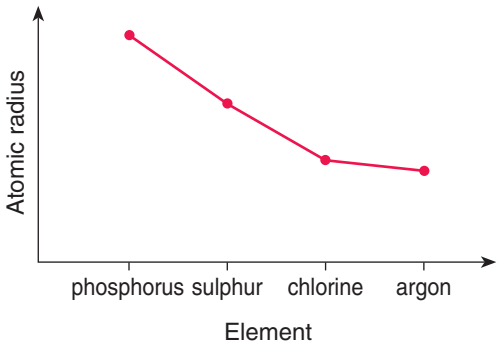
(Edexcel Advanced Subsidiary GCE, Unit 1, Jun. 2009, 23(a)–(c))

20 b) Strong covalent bonds exist between silicon atoms.
Van der Waals' forces exist between sulphur molecules.
Covalent bonds are much stronger than van der Waals' forces.
So, much more heat is needed to break the covalent bonds between silicon during melting.

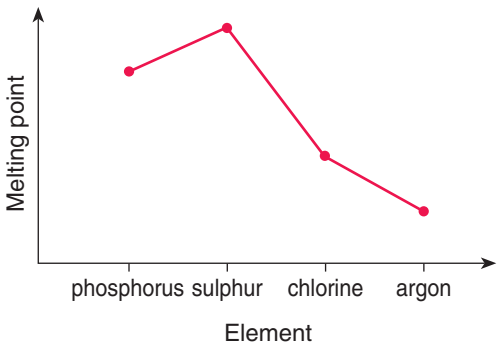
c) The melting point of an element depends on the strength of its inter-particle attractions.
The number of delocalized electrons available for metallic bonding increases in the order: Na < Mg < Al.
The size of the positive ion decreases in the order: Na > Mg > Al. The smaller is the ion, the closer the electrons are to the positive nucleus and the stronger is the attraction.
In addition, the degree of compactness of the packing of positive metallic ions increases in the order: Na < Mg < Al.
The strength of metallic bonds and thus the melting points of the elements increase in the order: Na < Mg < Al.

21 This question refers to the following Period 3 elements:
Refer to page T25.
phosphorus, sulphur, chlorine and argon

a) Sketch the trend in atomic radius of the elements. Account for the trend.



b) Sketch the trend in melting point of the elements. Account for the trend.



c) The electronegativity of an element represents the power of an atom of that element to attract a bonding pair of electrons towards itself in a molecule.

State and explain the trend in the electronegativity of phosphorus, sulphur and chlorine.

22 There is a link between the properties of the oxides of the Period 3 elements and their structure and bonding. The table below shows the melting points of the oxides of some Period 3 elements.

	Na_2O	SiO_2	P_4O_{10}
T_m (K)	1 548	1 883	573

a) In terms of crystal structure and bonding, explain in each case why the melting points of sodium oxide and silicon dioxide are high.

b) Predict whether the melting point of lithium oxide is higher than, the same as, or lower than the melting point of sodium oxide and explain your prediction.

c) Phosphorus pentoxide has a lower melting point than sodium oxide.

i) State the structure of and bonding in phosphorus pentoxide.

ii) Explain why the melting point of phosphorus pentoxide is low.

d) Separate samples of phosphorus pentoxide and sodium oxide were reacted with water.

In each case, predict the pH of the solution formed and write an equation for the reaction.

e) Write an equation for the reaction between Na_2O and P_4O_{10}. State the general type of reaction illustrated by this example.

(AQA Advanced GCE, Unit 5, Jan. 2011, 3)

22 a) Type of bonding of Na_2O: Ionic bonding
There are strong forces of attraction between the ions.
Type of bonding of SiO_2: Covalent bonding
There are strong covalent bonds between the atoms.
b) Prediction: Higher
Explanation: A lithium ion is smaller than a sodium ion.
A lithium ion can attract the oxide ions more strongly.
c) i) Type of structure: Simple molecular structure
Type of bonding: Covalent bonds exist between the phosphorus and oxygen atoms.
ii) Weak van der Waals' forces exist between the molecules.
Only very little heat is needed to separate the molecules during melting.
d) pH with P_4O_{10}: −1 to +2
Equation: $P_4O_{10}(s) + 6H_2O(l) \longrightarrow 4H_3PO_4(aq)$
pH with Na_2O: 12 to 14
Equation: $Na_2O(s) + H_2O(l) \longrightarrow 2NaOH(aq)$
e) Equation: $6Na_2O + P_4O_{10} \longrightarrow 4Na_3PO_4$
Reaction type: Acid-base

23 a) Using the following notations to complete the table below so as to provide information about the structure and acid-base property of the oxides of Period 3 elements.

IC: ionic crystal
CN: covalent network
SM: simple molecular structure
AC: acidic
BA: basic
AM: amphoteric

	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2
Structure					
Acid-base property					

b) By considering the trend of acid-base property and that of bonding of these oxides, state the relationship between the two trends.

c) Outline chemical tests to show how these oxides can be classified into acidic, basic and amphoteric.

(Communication mark will be awarded for this part.)

(HKDSE, Practice paper 1B, 2012, 9)

24 a) Write the electronic arrangement of a vanadium atom (Z = 23).

b) What is the oxidation state of vanadium in $VOBr_2$?

c) Write a balanced equation for the reaction between $VO_2^+(aq)$ and Zn(s) in an acidic medium to give $V^{2+}(aq)$ and $Zn^{2+}(aq)$.

(HKASLE, Paper 1, 2012, 2)

25 In the early 1800s, chemists had to learn the individual properties of all elements. Fortunately, they then developed ways of grouping the elements so that patterns of chemical and physical behaviour emerged. This led to our modern periodic table.

a) The Russian chemist Mendeléev organized the known elements into a pattern widely regarded as the first periodic table.

A version of Mendeléev's periodic table is shown below.

	Group I	**Group II**	**Group III**	**Group IV**	**Group V**	**Group VI**	**Group VII**	**Group VIII**
Period 1	H							
Period 2	Li	Be	B	C	N	O	F	
Period 3	Na	Mg	Al	Si	P	S	Cl	
Period 4	K Cu	Ca Zn	* *	Ti *	V As	Cr Se	Mn Br	Fe, Co, Ni
Period 5	Rb Ag	Sr Cd	Y In	Zr Sn	Nb Sb	Mo Te	* I	Ru, Rh, Pd

i) Why did Mendeléev leave gaps in his periodic table?
For elements yet to be discovered.

ii) How did later evidence from the scientific community support Mendeléev's decision to leave gaps in his periodic table?
New elements showing properties that fit in with the groups.

iii) Which group in the modern periodic table is missing from Mendeléev's table?
Noble gases

b) Before Mendeléev, John Newlands, another chemist, had been thinking on similar lines. He also identified patterns in the behaviour of elements. The table below shows how Newlands grouped the elements.

H	F	Cl	Co / Ni	Br	Pd	I	Pt / Ir
Li	Na	K	Cu	Rb	Ag	Cs	Tl
Gi	Mg	Ca	Zn	Sr	Cd	Ba / V	Pb
Bo	Al	Cr	Y	Ce / La	U	Ta	Th
C	Si	Ti	In	Zr	Sn	W	Hg
N	P	Mn	As	Di / Mo	Sb	Nb	Bi
O	S	Fe	Se	Ro / Ru	Te	Au	Os

i) Newlands and Mendeléev both put the elements in order of atomic mass.

What property is used to order the elements in the modern periodic table?
Atomic number

ii) Suggest why Newlands' arrangement was less useful than Mendeléev's arrangement.
Properties of some of the elements did not match up in Newlands' arrangement.

c) One physical property chemists examined was the melting point of the elements. This is tabulated below for the Period 2 elements.

Element	Li	Be	B	C (diamond)	N	O	F	Ne
Atomic number	3	4	5	6	7	8	9	10
Melting point (K)	453	1 553	2 349	3 773	63	54	53	24

i) Describe the pattern in melting point as you go across the period.
Goes up then goes down

ii) The change in melting point across the period can be explained in terms of the structure and bonding of the elements.
Bonding — metallic on the left changing to covalent when going right.
Structure — giant metallic structures at start; giant covalent structures in the middle; simple molecular structures on the right.

Describe the changes, both in the type of bonding and structure, as the period is crossed from left to right.

d) The electronic arrangement of an atom of an element is 2,8,8,2.

In which group and period of the modern periodic table is this element found?
Group II
Period 4

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, Jun. 2009, 2)